Language	ARC				MMLU			
	LlaMa2 7B	Gemma 7B	Mistral 7B	Qwen 7B	LlaMa2 7B	Gemma 7B	Mistral 7B	Qwen 7B
Chinese	27%	71%	57%	66%	32%	54%	37%	44%
German	27%	68%	63%	32%	25%	57%	47%	27%
French	31%	76%	59%	42%	24%	58%	48%	28%
Spanish	31%	77%	60%	46%	29%	56%	52%	33%
Italian	29%	77%	67%	44%	23%	56%	44%	32%
Kannada	24%	48%	27%	21%	21%	40%	22%	19%
Hindi	28%	60%	42%	22%	25%	45%	32%	23%
Armenian	19%	40%	36%	20%	20%	36%	30%	25%
Marathi	28%	46%	26%	25%	27%	42%	30%	26%
Telugu	30%	42%	30%	30%	24%	33%	34%	23%

Table 2: Performance on two inference tasks

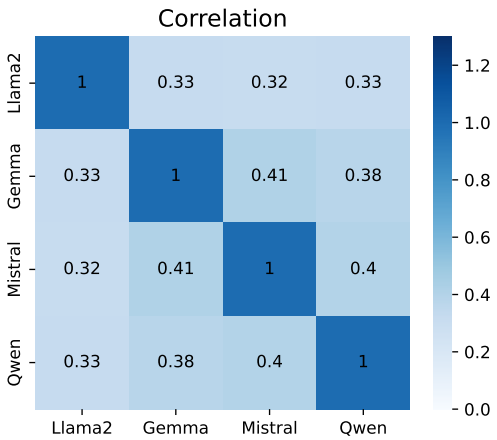

Figure 2: Rank correlation between different LLMs. This is calculated using metric introduced in Section 2.3. It shows high correlations across LLMs.

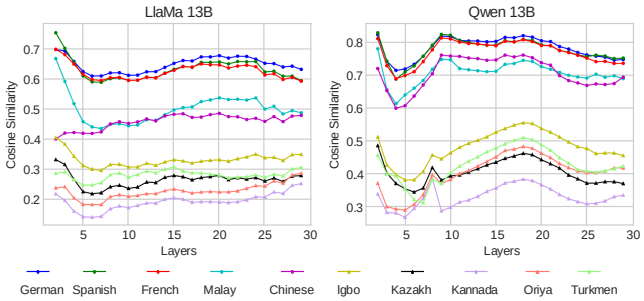

Figure 3: Cosine similarity scores across model layers for LlaMa2 13B and Qwen 13B. The graph shows results for the same 10 languages as in Figure 1. The findings reveal that the general trends observed in 7B models persist in these larger models.

Relationship to Ratio of Training Corpus? (RQ3)

In this section, we explore the relationship between the proportion of each language in the LlaMa2 pre-training corpus and their corresponding performance as measured by the similarity metric. According to the technical report of LlaMa2 (Touvron et al. 2023), we obtain the proportion of the pre-training corpus of some languages. Table 1 illustrates this relationship by listing a selection of languages with their proportion in the training data and their similarity scores relative to English. The table is divided into two parts: the left side lists high-resource languages with relatively higher proportions in the LlaMa2 pre-training corpus, and the right side lists low-resource languages with very low proportions ($\leq$ 0.01%). For example, German, with a proportion of 0.17%, has a high similarity score of 0.723, indicating strong performance in comparison to English. This trend suggests that languages with a higher proportion in the pre-training corpus tend to have higher similarity scores, reflecting better model performance. In contrast, low-resource languages like Kannada and Urdu, each with proportions of less than 0.01%, have much lower similarity scores (i.e., 0.236 and 0.275).

Correlation with Other Inference Tasks? (RQ4)

To more comprehensively reflect the performance of LLMs in various languages, we evaluate the multilingual reasoning ability of these LLMs.

We use MLMM-evaluation[2] as our benchmark dataset to evaluate LLMs' performances on reasoning tasks in various languages. The benchmark dataset can be used to evaluate the LLM across 26 different languages. It consists of three datasets: ARC (Clark et al. 2018), HellaSwag (Zellers et al. 2019), and MMLU (Hendrycks et al. 2020). We chose ARC and MMLU for evaluation, and both of them are multiple-choice datasets. The ARC dataset consists of 7,787 multiple-choice science questions drawn from a variety of sources. The MMLU dataset contains multiple-choice questions derived from diverse fields of knowledge. We selected five high-resource languages (Chinese, German, French, Spanish, Italian) and five low-resource languages (Kannada, Hindi, Armenian, Marathi, Telugu) for evaluation, randomly selected 100 samples from each language for 4-shot learning prediction, and used accuracy as the metric. The LLMs evaluated are consistent with Figure 1: LlaMa2 7B, Gemma 7B, Mistral 7B, and Qwen 7B.

[2] https://github.com/nlp-uoregon/mlmm-evaluation

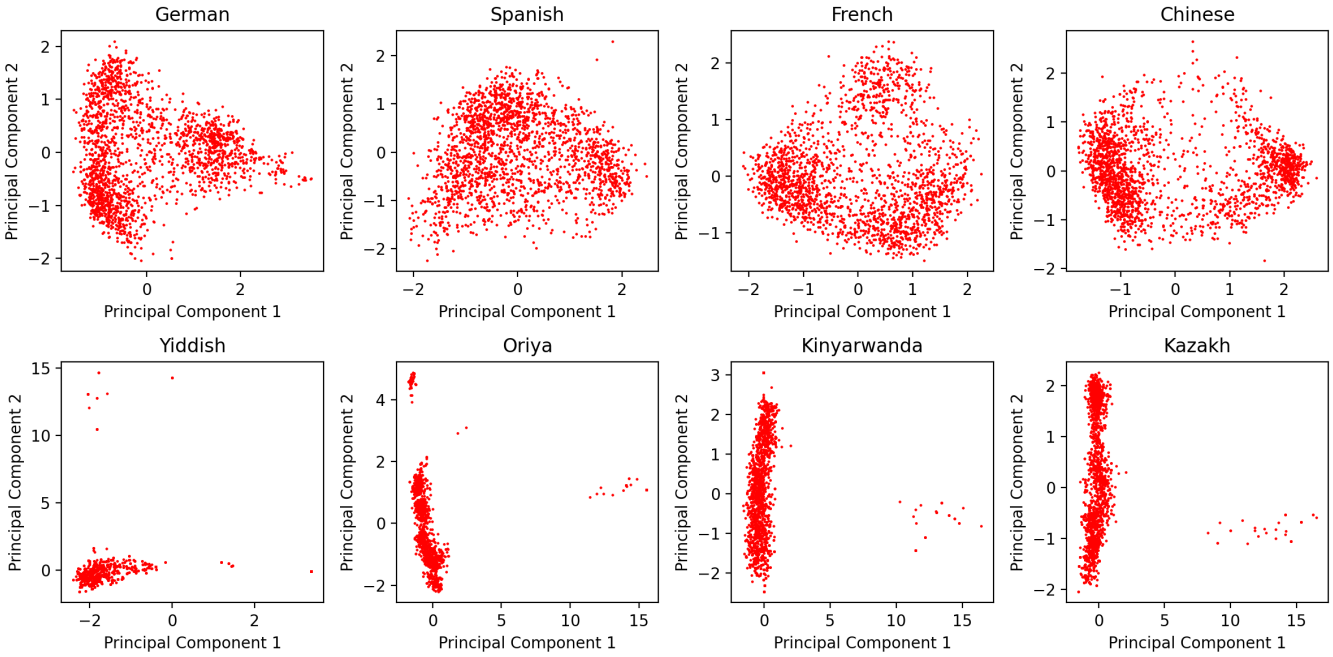

Figure 4: Visualization of the embedding space of Gemma 7B for eight languages. Four figures at the top are high-resource languages, Four figures at the bottom are low-resource languages.

High-High	Similarity Score	High-Low	Similarity Score	Low-Low	Similarity Score
English-German	0.72	German-Silesian	0.48	Azerbaijani-Turkmen	0.51
Italian-French	0.68	French-Erzya	0.35	Hungarian-Yiddish	0.24
German-French	0.67	Italian-Romany	0.32	Kab-SMT	0.36
French-Chinese	0.59	Italian-Uighur	0.27	Mari-Tatar	0.48

Table 3: Similarity score of different language pairs of Gemma 7B.

The predicted result is shown in Table 2. From the result, we find that for Gemma, Mistral, and Qwen, the performance of high-resource languages is significantly better than that of low-resource languages, and Gemma performs best. For the LlaMa2, the performance in all languages is generally not as good as the first three LLMs. This result shows that LLM reasoning ability in low-resource languages is worse than that in high-resource languages. This result proves that there are differences in performance between high-resource and low-resource languages in reasoning tasks, illustrating the effectiveness of the proposed cosine similarity metric.

Further Analysis of Proposed Metric

In the last section, we introduced and evaluated the Language Ranker, demonstrating its ability to quantify the multilingual capabilities of LLMs by comparing their internal representations against an English baseline. This provided a robust measure of how LLMs perform across different languages, especially highlighting the disparities between high-resource and low-resource languages.

Building on these insights, in this section, we delve deeper into the proposed metric to explore its credibility and reliability further. Specifically, we aim to answer the following questions in the following three subsections: RQ5: Is choosing English as the benchmark a wise choice? RQ6: What does the subspace of each language look like? RQ7: Is choosing cosine similarity a wise choice?

Why Using English as Baseline? (RQ5)

In the above sections, we choose English as a baseline. This is based on the a priori assumption that low-resource languages generally perform worse than high-resource languages. But if we choose other high-resource languages as baselines, will we get the same performance? In other words, how can we ensure that our metric is not affected by the English language itself? To answer this question, we divided our probing datasets into three types: High Resource-High Resource (**H-H**), High Resource-Low Resource (**H-L**), and Low Resource-Low Resource (**L-L**). To fulfill our requirement, we utilize Tatoeba-Challenge (Tiedemann 2020) as our dataset instead of opus-100 because the latter is an English-centric dataset which means there is no Low Resource-Low Resource language pair. Tatoeba-Challenge is a challenge set for machine translation that contains 32G translation units in 2,539 bitexts. The whole data set covers 487 languages linked to each other in 4,024 language pairs. We select four language pairs for each group,

High-High	Similarity Score	High-Low	Similarity Score	Low-Low	Similarity Score
English-German	0.72	German-Silesian	0.44	Azerbaijani-Turkmen	0.51
Italian-French	0.69	French-Erzya	0.31	Hungarian-Yiddish	0.19
German-French	0.68	Italian-Romany	0.15	Kab-SMT	0.40
French-Chinese	0.56	Italian-Uighur	0.20	Mari-Tatar	0.42

Table 4: Similarity score of different language pairs of LlaMa2 7B.

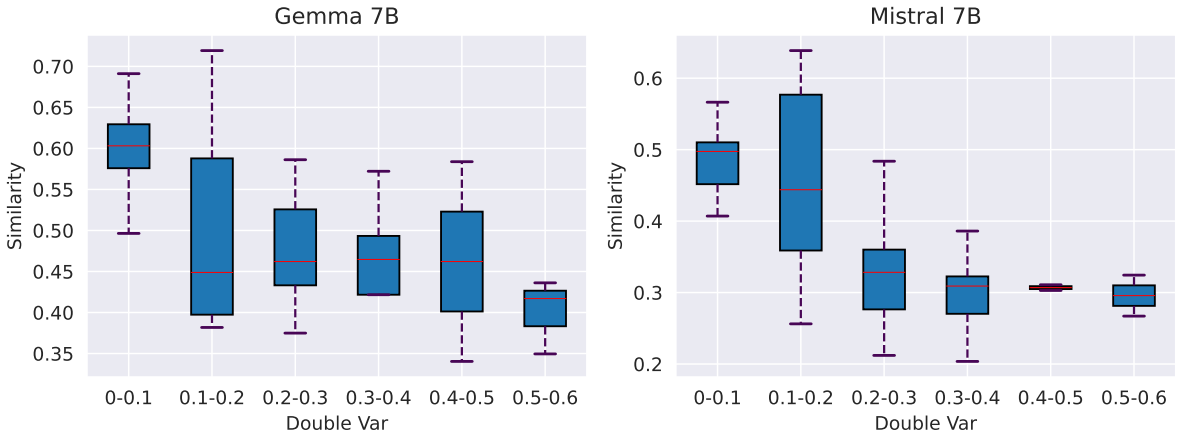

Figure 5: Box plot of the relationship between the double variance and similarity of Gemma 7B and Mistral 7B.

English-German (en-de), English-French (en-fr), German-French (de-fr), and Italian-German (it-de) represent **H-H**; German - Silesian (de-szl), French-Erzya (fr-myv), Italian - Romany (it-ro) and Italian - Uighur (it-uig) represent **H-L**; Azerbaijani-Turkmen (az-tr), Hungarian-Yiddish (hu-yi), Kabyle-Standard Moroccan Tamazight (kab-SMT) and Mari-Tatar (ma-ta) represent **L-L**. The results are shown in Table 3 and Table 4.

From the results, we can observe that the score of High-High is higher than the score of High-Low and Low-Low universally. An obvious inference is that the distribution of high-resource languages is relatively close to each other, while the distribution of low-resource languages varies greatly, neither being close to each other nor to high-resource languages. Therefore, the distribution of high-resource languages is relatively consistent, while the distribution of low-resource languages varies greatly. Choosing English as the baseline is a convenient choice. Thus, we can also choose other high-resource languages such as German and French as the baseline.

Deeper Analysis of the Embedding Space (RQ6)

We explained why we chose English as the baseline in the above section. Choosing English is actually choosing a high-resource language as our baseline. A question naturally arises: How can we make sure that the performance of high-resource languages is better than the performance of low-resource languages? The result of Table 2 has confirmed the answer to the question by the reasoning task. To answer this question more deeply, we need to analyze the distribution of the embedding of different languages. As is shown in Figure 4, the top four sub-figures are embedding spaces of high-resource languages, while the bottom four sub-figures are embedding spaces of low-resource languages. It is obvious that high-resource languages are more evenly distributed throughout the space, while low-resource languages are more narrowly distributed, and compressed into a near straight line. Therefore, the performance of low-resource languages is worse than that of high-resource languages, which means it is suitable to choose a high-resource language like English as the baseline.

Why Using Cosine Similarity? (RQ7)

Recent research (Steck, Ekanadham, and Kallus 2024) has shown that cosine similarity is not always a reliable metric. Inspired by Section , the quality of the performance of various languages can be clearly judged from the subspace distribution. Therefore, we decided to quantify the performance of these languages from a distribution perspective.

Back to Figure 4, the projection distributions of high-resource languages in different directions are relatively consistent, while the distribution of low-resource languages is compressed approximately into a straight line. The projection distributions outside the straight line, such as the projections perpendicular to the straight line, are crowded in a smaller area. This suggests that we can use the projection variance to approximately measure the quality of the distribution.

According to PCA, we assume the embedding vectors are $\{X_i\}_{i=1}^{n}$(being centralized), the projection direction is ω,